Problem Set 2

Due date: 9/22/2025

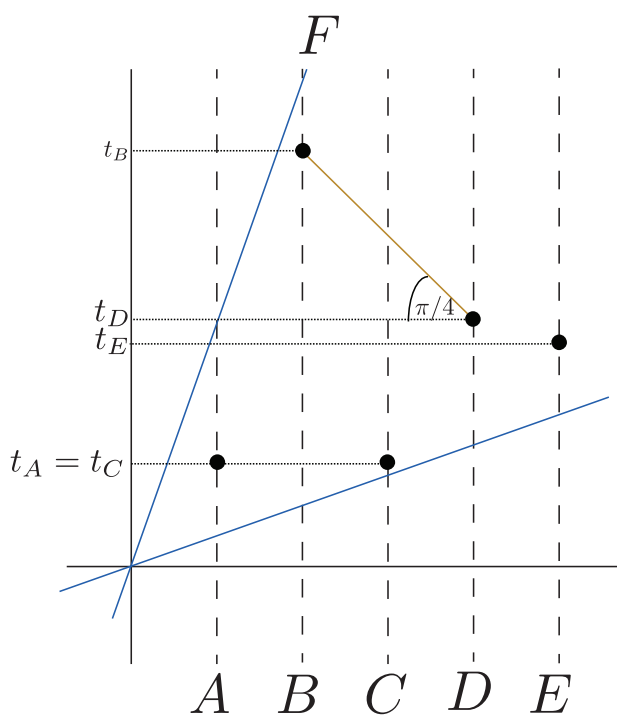
1. *(20 points)* Using natural units:

   a. *(10 points)* Convert the following quantities to natural units: 5 Joules of energy, a density of $10kg/m^3$, an acceleration of $3m/s^2$.

   b. *(10 points)* Convert the following quantities from natural units to SI: a velocity of 5, and a time of 3.

2. *(15 points)* Flesh out the algebra with factors of c and show using a Taylor expansion for small velocities that the Lorentz transformations reduce to Galilean transformations.

3. *(20 points)* Alice, Bob, Charlie, Diana, and Eli are simultaneously at rest in reference frame R, spaced out in the x direction as drawn in the figure below. Each of $A - E$ are holding a flashlight, which they turn on at the different times t_A, t_B, t_C, t_D, t_E marked in the diagram below.



   a. *(5 points)* What is the order in which the light from the flashlights reaches an observer sitting at rest at the spatial origin?

   b. *(5 points)* Fred is moving at some velocity v in the $+x$ direction (blue line in the diagram). In Fred's rest frame, what is the order in which the flashlights turn on? Justify your answer through a spacetime diagram. I strongly suggest you use a ruler!

   c. *(10 points)* What is the order in which the light from the flashlights reaches Fred?

4. *(20 points)* You and a friend decide to race one another, but the racetrack is too narrow for both of you. Instead you decide to run in opposite directions starting halfway between two finish lines in the racetrack's rest frame. The referee, who is at rest with respect to the racetrack, pronounces the race to be a tie. Assuming you and your friend both moved with constant velocity throughout the race, do you both agree with the referee or not? Make sure to justify your answer, either with equations or with a spacetime diagram, to receive credit!

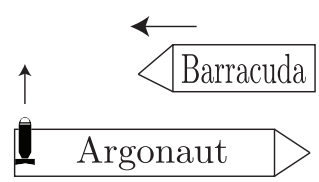
5. *(20 points)* You and your friend decide to race each other again, this time starting on opposite sides of a finish line and equidistant from the finish line in the racetrack rest frame. The referee, who is again at rest with respect to the racetrack, sees that you and your friend both started at the same time and finished at the same time and that it must therefore be a tie. Do you agree that you started at the same time? Do you agree that you finished at the same time? Again, justify your answer with either equations or a spacetime diagram.

6. *(25 points)* A rod of length L is at rest in R, with one end at the origin and one end at $x = +L$. According to the frame R', R moves in the $+x$ direction with speed v. R and R' coincide at $t = 0$ (i.e. there is no rigid translation between them). At $t = 0$, which coincides with $t' = 0$, a flash of light is emitted. Compute the coordinates of the event at which this flash of light reaches the other end of the rod in both R and R'.

7. *(35 points)* The two submarines USS Barracuda and USS Argonaut are traveling towards one another. The relative velocity between them is v. For purposes of this problem you may ignore water friction. The proper length of each submarine is L_0. The submarines are slightly offset so one is deeper than the other.

   As part of a military exercise, USS Argonaut fires a torpedo vertically towards the USS Barracuda as soon as the tail of the Barracuda passes the head of the Argonaut in the Argonaut rest frame:

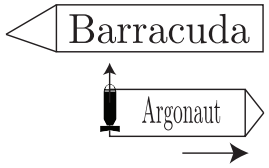
   Argonaut rest frame



   (Assume that the torpedo is very fast and the offset is very small so we may approximate the torpedo as traveling through the offset instantaneously.) In the Argonaut rest frame (let's call it frame A), the torpedo must leave the Barracuda unharmed since the length of the Barracuda is contracted. However, in the Barracuda rest frame, the Argonaut is contracted, suggesting that the figure on the next page is the correct scenario.

   To determine the if Barracuda gets hit, determine the coordinates of the following in both the A and B frames:

Barracuda rest frame



a. the head of the Argonaut and tail of the Barracuda are aligned in the Argonaut rest frame (frame A).

b. the torpedo is fired.

c. the tail of the Argonaut and the head of the Barracuda are aligned.

Does the Barracuda get hit or not? Do the sailors on the Argonaut agree with the sailors on the Barracuda?